

NGUYEN HOAI NAM

FRESHER UNITY GAME DEVELOPER

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PROFESSIONAL SUMMARY

Unity and C# game developer and final-year **Information Technology** student at UTH. I build gameplay systems and Unity editor tools, and I optimize by measurement: in my farming game I cut main-thread time from **272 ms to 41 ms** using the Unity Profiler and an in-game PerfHud I wrote. As team leader of **"Three Musketeers"**, a 3-person competitive programming team, I designed the strategy of our C++ game AI, which ranked **86th of 1,570** in Nexon's NYPC 2026.

CORE TECHNICAL SKILLS

Engine & Languages:	Unity 6 (URP 2D), C#, C++
Game Systems & Math:	ScriptableObject data, state machines, grid systems, object pooling, Input System, uGUI / TMP, 2D physics (Rigidbody2D, casts), vector math
Profiling & Optimization:	Unity Profiler (CustomSampler, ProfilerRecorder), Frame Debugger, draw call batching, Sprite Atlas, Tilemap chunking, GC-free loops
Tools & Workflow:	Unity Editor scripting (MenuItem tools, one-click build tool, Sprite Editor data provider API), unit tests for pure C# logic, Git / GitHub
Platforms & Other:	Android APK (ARM64), WebGL, Windows Algorithms (Dijkstra, simulation) English: TOEIC 600

COMPETITIVE PROGRAMMING

Nexon Youth Programming Challenge (NYPC 2026), Master Track	Jun 2026 – Aug 2026
Team Three Musketeers (3 members) • Team Leader & Strategy Designer • C++ strategy AI for "NEXT NATION" • 7 NYPC badges	
Led a 3-person team: studied the rules and designed the strategy myself: the per-turn loop that moves every unit step by step, flank and threat analysis, and when to expand, defend or attack. Split the supporting functions between 2 teammates (shortest-path cache, combat simulator, gold-safe commands) and merged them into one bot. Improved from data: reviewed 7 lost matches with the team and shipped 5 targeted fixes; the new version won 8/8 pretest matches. Results: final rank 86 / 1,570 (Top 5%), Round 1 win rate 84%, Top 1 and Top 10 in preliminary tests, Top 10–30 in live qualification; under 1 ms per turn vs a 100 ms limit, zero timeouts.	

GAME PROJECTS

Cooking-Game-2D (FarmHn) Solo Developer	GitHub ↗ Itch.io ↗ Mar 2026 – Present
Isometric farming and cooking sim • Unity 6.3 (URP 2D), C# • 658 C# scripts, 4 scenes • Android APK (128 MB) and WebGL, both playable on Android phones	
Game features: isometric grid with building placement preview, crops and livestock, feed mill, kitchen cooking, market stall and order board, shop and warehouse, trains and tourist boats, land expansion, save system, day-night cycle and weather, tutorial, Vietnamese / English localization. - Profiled a ~4 FPS build with Unity Profiler, CustomSampler and my in-game PerfHud; traced ~60% of main-thread time to uncached Type.GetType() reflection in per-frame input checks. Cached lookups, moved the cheap touch check first: main thread 272 → 41 ms, FPS 3.9 → 30–40, 1% low 0.7 → 12–15 (PC, PerfHud). - Cut draw calls 81% (1,652 → 309): Tilemap Chunk mode for 22,000+ hidden-layer tiles and one Sprite Atlas for 64 decor sprites; kept visible layers Individual after tracing seams to Tight sprite meshes. - Removed 5 permanent FindObjectsByType sweeps over 1,860 GameObjects, de-synced 27 same-frame string allocators, and built a Light2D view culler (520 → 271 active lights).	

- Fixed build-only bugs the Editor hid: `CultureInfo("vi-VN")` crashed the Shop on stripped IL2CPP builds, `#if UNITY_EDITOR` popups were missing in builds, and 73 duplicate keys blocked the 1,677-key localization table.
- Built tools: runtime **PerfHud** (F3) on `ProfilerRecorder`, ffmpeg frame analysis of phone videos (17 FPS Android baseline), and editor tools that slice sprite sheets into animations and prefabs with stable sprite IDs.
- **Measured every change:** the first 23-file "best practices" pass, done without measuring, made the game slower. I then fixed one bottleneck at a time and re-measured after each change, over many rounds, until the game held 30–40 FPS.

SkyBound (Super Skybound Adventure) | Solo Developer

[GitHub ↗](#)
[Itch.io ↗](#)

Aug 29 – Sep 9, 2026

2D mobile platformer • Unity 6.3 (URP 2D), C# • 3 stages plus Home and Victory scenes • Android APK (42 MB, ARM64) and WebGL (21 MB) • Kenney art assets

- **Game features:** 3 stages with key-locked doors, coins, gems, heart pickups, spring pads, spikes and fall zones; a 3-2-1 countdown, game over and victory flow; on-screen touch controls.
- **Character controller:** `Rigidbody2D` movement with acceleration 60 and deceleration 50, coyote time and jump buffer (0.15 s each), variable jump height (fall gravity $\times 2.5$, short hop $\times 2$), `BoxCast` ground check, and continuous collision detection against tunneling.
- **Health and feedback:** 5 hearts, knockback, 1.5 s invincibility with hit flash; animation driven by controller events (`OnJump`, `OnLand`, `OnMove`).
- **Mobile controls and camera:** touch buttons on pointer down / up / exit, so a finger sliding off a button releases it; `SmoothDamp` camera with look-ahead offset and level bounds.
- **Editor tools:** a one-click build tool (Android APK on ARM64, WebGL zipped for itch.io, Windows) and a level setup tool that imports 65 tiles and 18 items into a Tile Palette and creates 5 tilemap layers (background, ground, platforms, hazards, props).

EDUCATION

University of Transport Ho Chi Minh City (UTH) | B.Sc. Information Technology Oct 2022 – Dec 2026 (expected)